



A+ LAB SERIES V5

Lab 5: Command Line – Windows and Linux

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Material in this Lab Aligns to the Following	
CompTIA A+ (220-1201) Exam Objectives	5.5: Given a scenario, troubleshoot network issues.
CompTIA A+ (220-1202) Exam Objectives	1.3: Compare and contrast basic features of Microsoft Windows editions. 1.5: Given a scenario, use the appropriate Microsoft command-line tools. 1.9: Identify common features and tools of the Linux client/desktop operating system. 2.7: Given a scenario, apply workstation security options and hardening techniques.

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Introduction

In this lab exercise, participants will compare command-line interface (CLI) operations across two operating systems: Windows and Linux. The CLI is a text-based user interface, commonly called a shell, that enables direct interaction between the user and the operating system to execute system commands and tasks. Windows uses its proprietary shell, notably Command Prompt (cmd.exe) or PowerShell, whereas Linux distributions typically use shells such as Bash (Bourne Again Shell), Zsh (Z Shell), and others. This exploration will involve typing and executing commands specific to each shell, highlighting the differences in syntax, functionality, and user interaction paradigms between these environments.

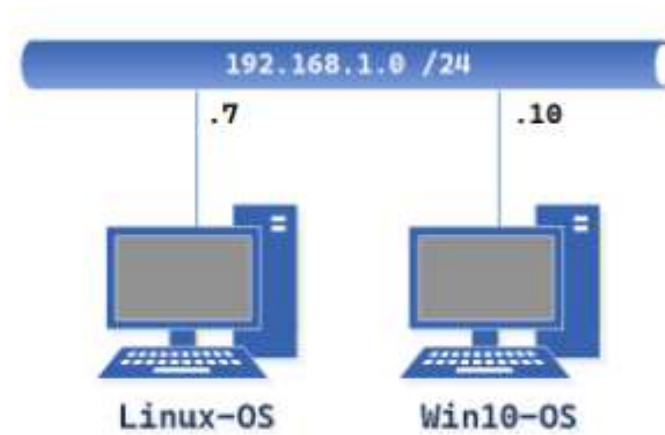
Objective

The objective of this lab session is to acquire proficiency in navigating and executing fundamental commands within the command-line interfaces of both the Windows Command Prompt and the Ubuntu terminal. Participants will engage in hands-on exercises designed to familiarize themselves with the essential commands necessary for basic system navigation, file management, and the execution of system utilities. This dual-platform approach will not only enhance the understanding of command-line operations specific to each environment but also foster an appreciation for the underlying principles of operating system interaction through textual commands and review key concepts related to Windows and Linux CLI operations.

In this lab, we will be performing the following tasks:

- Navigate the Windows CLI.
- View file contents, create directories, and create files with the Windows CLI.
- Navigate and utilize the Windows CLI with Administrative Privileges.
- Navigate and utilize the Linux CLI with Administrative Privileges.
- View file contents, create directories, and create files with the Linux CLI.
- Change between drives using CLI and locate files using wildcards.

Lab Topology



Lab Settings

The information in the table below will be needed in order to complete the lab. The task sections below provide details on the use of this information.

Virtual Machine	IP Address	Account (if needed)	Password (if needed)
Win10-OS	192.168.1.10 / 24	sysadmin	NDGLabpass123!
Linux-OS	192.168.1.7 / 24	sysadmin	NDGLabpass123!

1 Navigate the Command Line Interface in Windows

The Windows Command Line Interface (CLI), commonly known as Command Prompt or `cmd.exe`, is a command-line interpreter available on most Windows operating systems. It provides users with a text-based interface to interact with the operating system, allowing them to execute commands, scripts, and batch files. Here's a brief overview of some fundamental aspects and commands in the Windows CLI:

- **Command Prompt (`cmd.exe`):** This is the traditional command-line interface in Windows, offering a straightforward way to perform tasks like file management, system administration, and network operations.
- **PowerShell:** A more advanced shell and scripting language built on the .NET Framework, designed for task automation and configuration management. While it can run traditional CMD commands, it also introduces more powerful cmdlets and scripting capabilities.

1.1 Using the CLI in Windows 10

The *Command Prompt* in *Windows* provides access to many commands that can be run from the *GUI*, but it is important to understand *CLI* basics when the *Windows GUI* may be unavailable. These commands perform operating system tasks from the command line instead of the more commonly used Windows GUI.

1.1.1 Viewing Contents in a Directory in Windows 10

1. Open a console connection to the **Win10-OS** system.

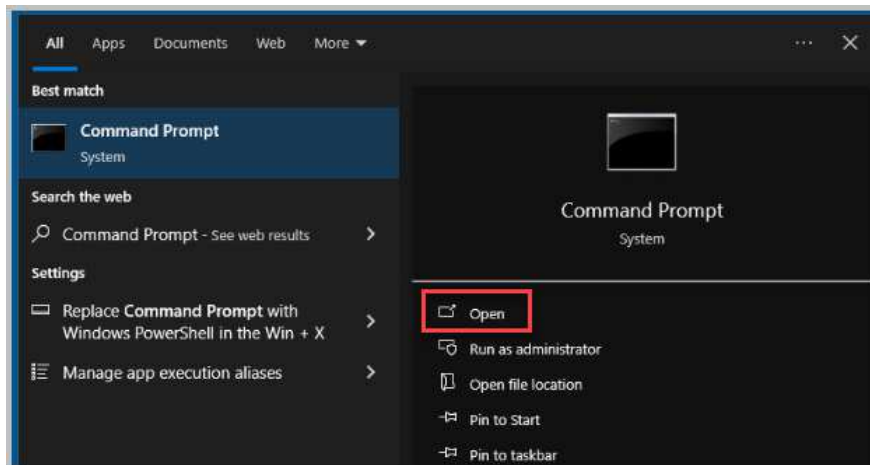


To launch the console for a lab device, you can click on the respective tab in the navigation bar at the top or navigate to the *Topology* section and click the lab device's corresponding image.

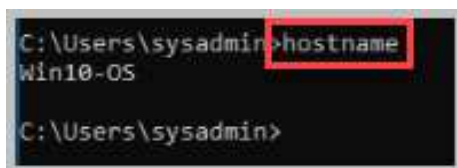
2. To begin interacting with the command line interface, you will need to open a *Command Prompt* shell. In the taskbar of the Win10-OS computer search bar, type **Command Prompt**.



3. Select **Open** to open the Command Prompt Window.

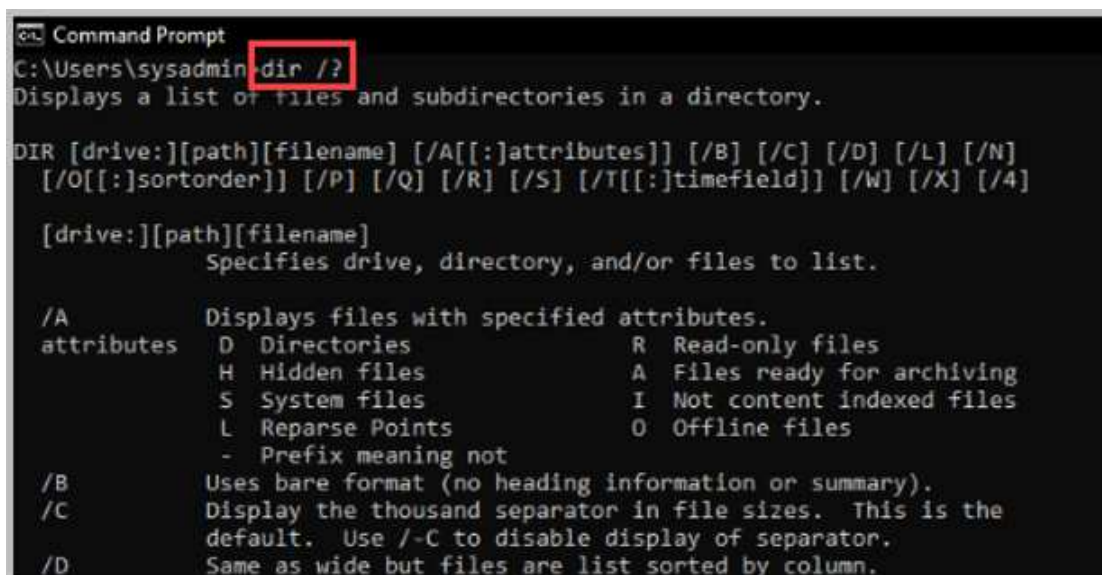


4. In the Windows 10 Command Prompt, knowing the name of the computer you're working on can be quite useful. To display the computer's hostname, simply enter `hostname` in the Command Prompt window. This command will immediately display your computer's name.



5. Click inside the shell and type the commands below, followed by pressing the **Enter** key. The commands below will help show you the `dir` command's parameters. Both commands accomplish the same objective of receiving help.

```
dir /?
```



Or...

help dir

```
C:\Users\sysadmin> help dir
Displays a list of files and subdirectories in a directory.

DIR [drive:][path][filename] [/A[:attributes]] [/B] [/C] [/D] [/L] [/N]
  [/O[:sortorder]] [/P] [/Q] [/R] [/S] [/T[:timefield]] [/W] [/X] [/4]

[drive:][path][filename]
    Specifies drive, directory, and/or files to list.

/A      Displays files with specified attributes.
attributes  D Directories          R Read-only files
             H Hidden files        A Files ready for archiving
             S System files        I Not content indexed files
             L Reparse Points      O Offline files
             - Prefix meaning not

/B      Uses bare format (no heading information or summary).
/C      Display the thousand separator in file sizes. This is the
        default. Use /-C to disable display of separator.
/D      Same as /B but files are list sorted by column.
/L      Uses lowercase.
/N      New long list format where filenames are on the far right.
/O      List by files in sorted order.
sortorder  N By name (alphabetic)    S By size (smallest first)
            E By extension (alphabetic) D By date/time (oldest first)
            G Group directories first - Prefix to reverse order

/P      Pauses after each screenful of information.
/Q      Display the owner of the file.
/R      Display alternate data streams of the file.
/S      Displays files in specified directory and all subdirectories.
```



When typing commands in the command prompt, typically at the end of every typed command you press the **Enter** key to initiate the command. Also, if you accidentally misspelled a command, as a shortcut, you could press the **up arrow** to bring back the previous commands you've entered.



To get help at any place in the command prompt, type **help** and a list of all available commands will display. If you need help with a specific command type **<command> /?**.

6. Type the command below and then press the **Enter** key. The *dir* command displays a list of the directories and files in the current directory. Notice that you are in the *C:\Users\sysadmin* directory.

dir

```
C:\Users\sysadmin>dir
Volume in drive C has no label.
Volume Serial Number is F2DC-9378

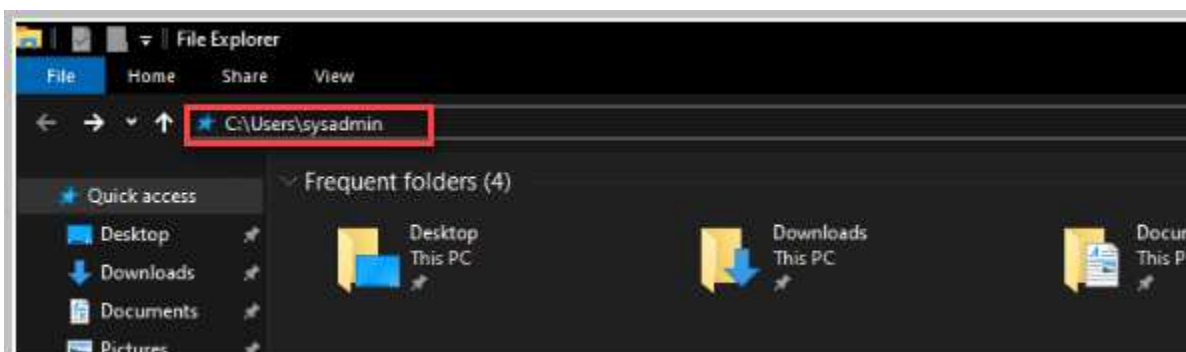
Directory of C:\Users\sysadmin

05/20/2024  04:10 PM  <DIR>          .
05/20/2024  04:10 PM  <DIR>          ..
03/23/2023  05:04 AM  <DIR>          3D Objects
03/23/2023  05:04 AM  <DIR>          Contacts
06/28/2023  07:09 PM  <DIR>          Desktop
06/27/2023  05:31 PM  <DIR>          Documents
06/28/2023  07:04 PM  <DIR>          Downloads
03/23/2023  05:04 AM  <DIR>          Favorites
03/23/2023  05:04 AM  <DIR>          Links
03/23/2023  05:04 AM  <DIR>          Music
05/20/2024  04:10 PM             1,310,720 NTUSER.bak
08/01/2024  03:26 PM             1,310,720 NTUSER.DAT
03/23/2023  03:54 AM  <DIR>          OneDrive
05/17/2023  04:45 PM  <DIR>          Pictures
03/23/2023  05:04 AM  <DIR>          Saved Games
03/23/2023  05:04 AM  <DIR>          Searches
03/31/2023  03:47 AM  <DIR>          Videos
                2 File(s)              2,621,440 bytes
               15 Dir(s) 13,968,957,440 bytes free
```

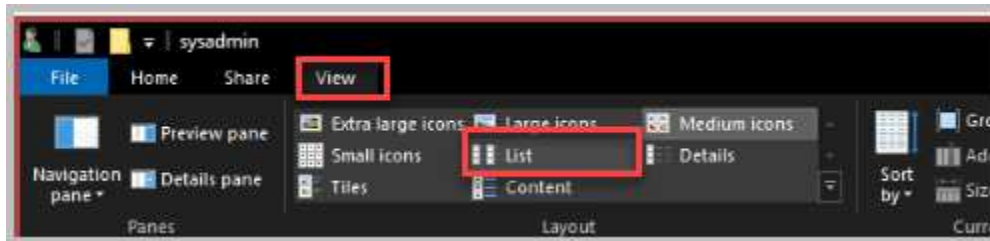
7. To compare the results in the shell with the graphical user interface, click on the **File Explorer** icon in the taskbar.



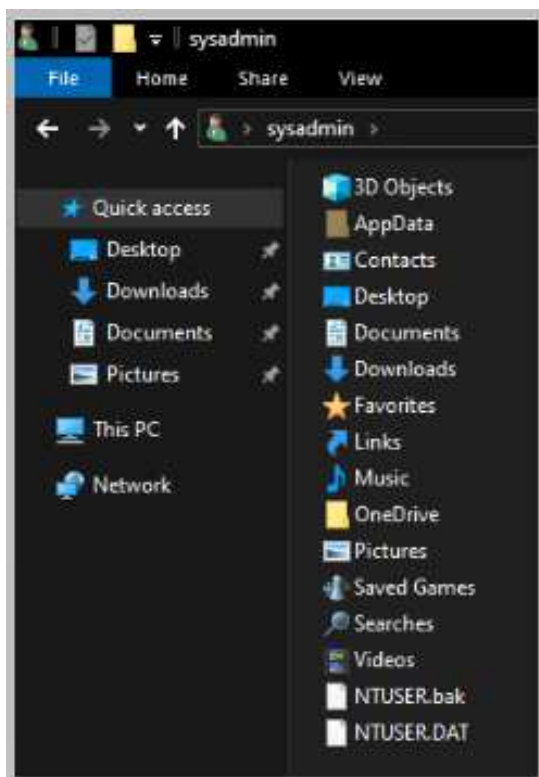
8. In the *File Explorer* window, type *C:\Users\sysadmin* in the search field, followed by pressing the **Enter** key in the address field.



- Change the view option for the sysadmin folder by selecting **View** on the menu bar. Select the **List** option to change the folder view.



- Notice the folders displayed here and compare them to the `dir` command from the *Command Prompt*. The same folders are displayed, as this is just another way of navigating through the same system.

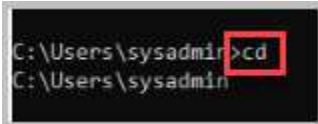


- Leave the *Win10-OS* window open to continue with the next task.

1.1.2 Working with Directories

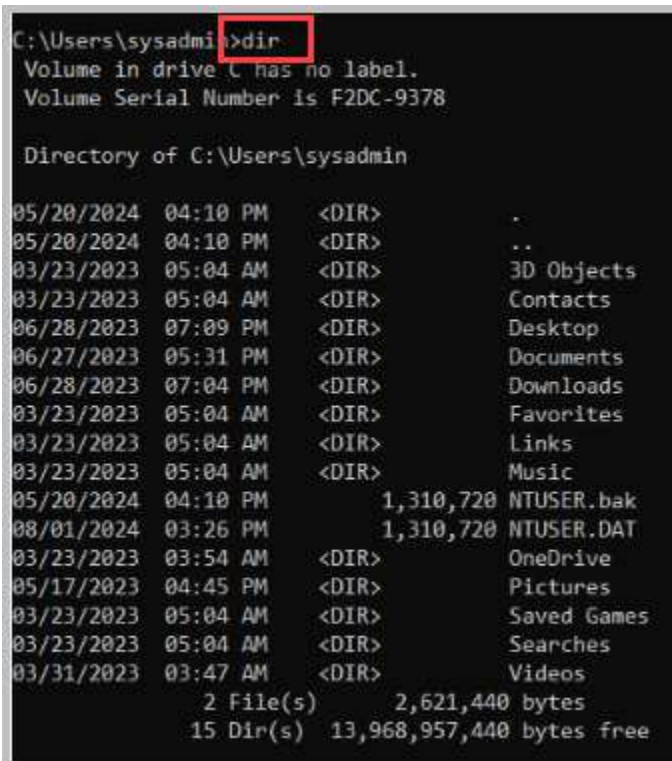
When interacting with the CLI, the prompt indicates the file system you are working in. The commands apply to the files in the current location unless you specify otherwise. It is very important to know the directory you are currently working in.

- Change focus back to the **command prompt**.
- In the command prompt window, type the command below, followed by pressing the **Enter** key to show the current working directory. Notice you are in the `C:\User\sysadmin` directory.

`cd`

```
C:\Users\sysadmin>cd
C:\Users\sysadmin
```

3. Type the command below and then press **Enter**. Notice the directory called *Documents*. Let's try to navigate to this directory.

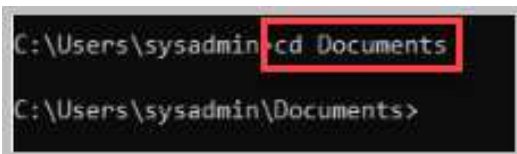
`dir`

```
C:\Users\sysadmin>dir
Volume in drive C has no label.
Volume Serial Number is F2DC-9378

Directory of C:\Users\sysadmin

05/20/2024  04:10 PM    <DIR>          .
05/20/2024  04:10 PM    <DIR>          ..
03/23/2023  05:04 AM    <DIR>          3D Objects
03/23/2023  05:04 AM    <DIR>          Contacts
06/28/2023  07:09 PM    <DIR>          Desktop
06/27/2023  05:31 PM    <DIR>          Documents
06/28/2023  07:04 PM    <DIR>          Downloads
03/23/2023  05:04 AM    <DIR>          Favorites
03/23/2023  05:04 AM    <DIR>          Links
03/23/2023  05:04 AM    <DIR>          Music
05/20/2024  04:10 PM             1,310,720 NTUSER.bak
08/01/2024  03:26 PM             1,310,720 NTUSER.DAT
03/23/2023  03:54 AM    <DIR>          OneDrive
05/17/2023  04:45 PM    <DIR>          Pictures
03/23/2023  05:04 AM    <DIR>          Saved Games
03/23/2023  05:04 AM    <DIR>          Searches
03/31/2023  03:47 AM    <DIR>          Videos
                2 File(s)          2,621,440 bytes
               15 Dir(s)  13,968,957,440 bytes free
```

4. Type the command below and press the **Enter** key. Notice that the prompt changes to *C:\Users\sysadmin\Documents*.

`cd Documents`

```
C:\Users\sysadmin>cd Documents
C:\Users\sysadmin\Documents>
```

5. Enter the `cd` command to show the current working directory and verify that you are in the *Documents* folder for the *sysadmin* user.

`cd`

```
C:\Users\sysadmin\Documents>cd
C:\Users\sysadmin\Documents
C:\Users\sysadmin\Documents>
```

6. Enter the `dir` command to view the contents of the current directory. Notice the `."`, which represents the current folder (*Documents*), and the `.."`, which represents the parent folder (*sysadmin*). Also, notice that there is a folder called *WindowsPowerShell*.

`dir`

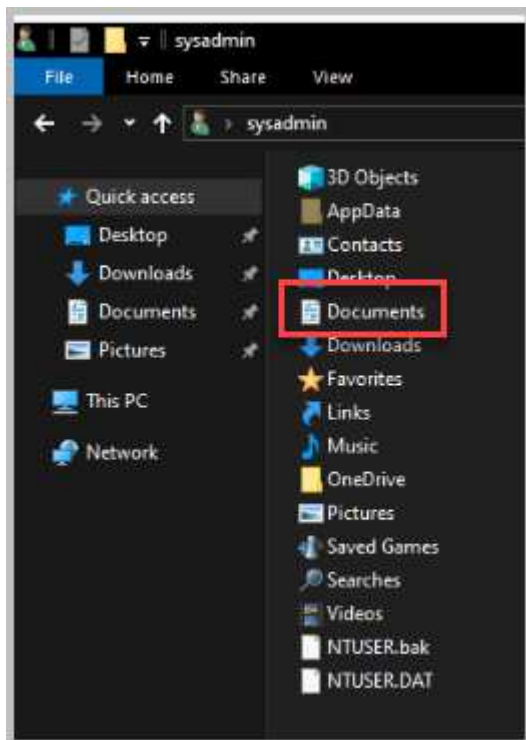
```
C:\Users\sysadmin\Documents>dir
Volume in drive C has no label.
Volume Serial Number is F2DC-9378

Directory of C:\Users\sysadmin\Documents

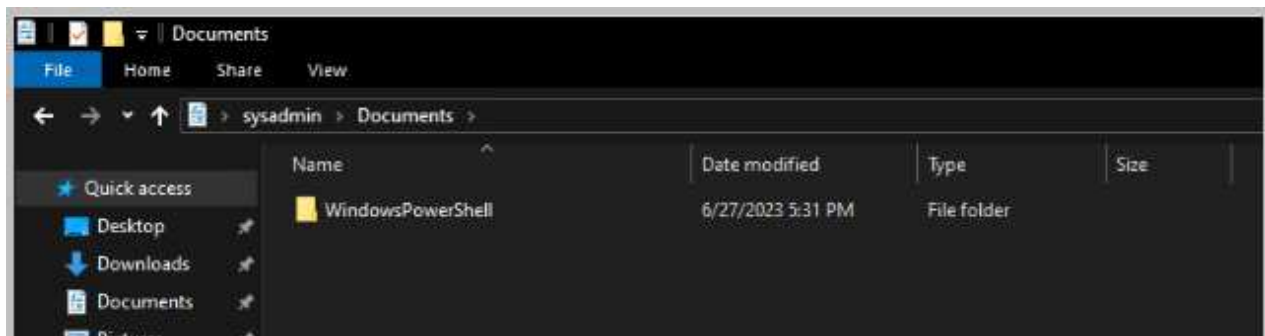
06/27/2023  05:31 PM    <DIR>          .
06/27/2023  05:31 PM    <DIR>          ..
06/27/2023  05:31 PM    <DIR>          WindowsPowerShell
               0 File(s)                0 bytes
               3 Dir(s) 13,968,154,624 bytes free

C:\Users\sysadmin\Documents>
```

7. Change focus to the **File Explorer** window and double-click on the **Documents** folder.

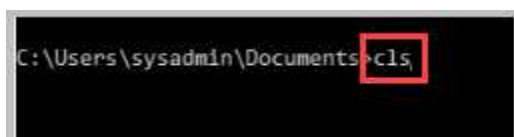


8. Compare the results from the command prompt. The same *WindowsPowerShell* folder is seen.



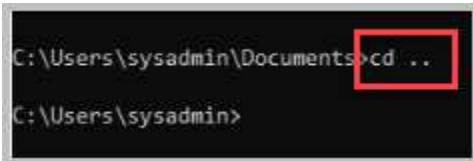
9. Change focus to the command prompt and enter the command below, followed by pressing the **Enter** key to clear the shell. This command is useful when non-pertinent information is filling up the shell.

```
cls
```



10. Type the command below, followed by pressing the **Enter** key to move back to the directory above the one in which you are currently working.

```
cd ..
```



```
C:\Users\sysadmin\Documents>cd ..  
C:\Users\sysadmin>
```

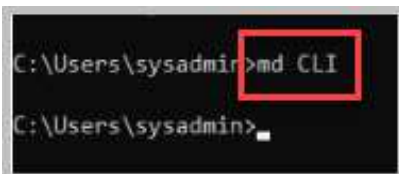
11. Notice you are back in the *C:\Users\sysadmin* directory, or, in other words, the *sysadmin* folder.
12. Leave the Win10-OS computer with the command prompt open to continue with the next task.

1.1.3 Creating Directories and Files

In this task, you will create new directories.

1. In the command prompt, type the command below, followed by pressing the **Enter** key to create a new directory called *CLI*.

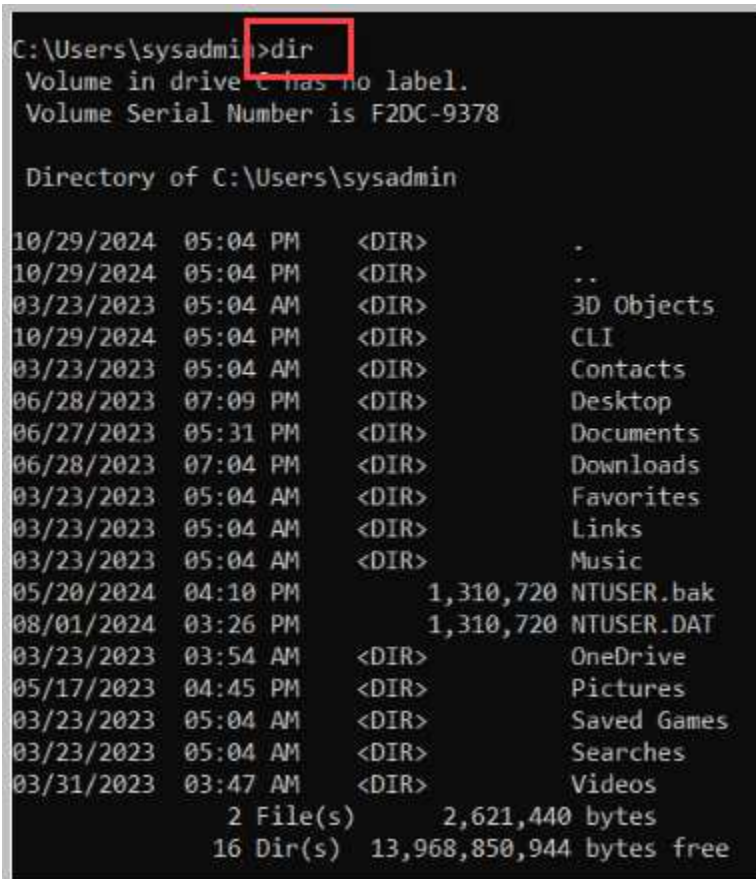
```
md CLI
```



```
C:\Users\sysadmin>md CLI  
C:\Users\sysadmin>
```

2. Verify that the new directory exists by listing your directory. Enter the **dir** command.

```
dir
```



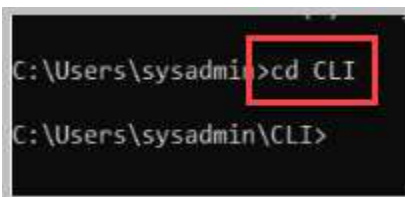
```
C:\Users\sysadmin>dir
Volume in drive C has no label.
Volume Serial Number is F2DC-9378

Directory of C:\Users\sysadmin

10/29/2024  05:04 PM    <DIR>          .
10/29/2024  05:04 PM    <DIR>          ..
03/23/2023  05:04 AM    <DIR>          3D Objects
10/29/2024  05:04 PM    <DIR>          CLI
03/23/2023  05:04 AM    <DIR>          Contacts
06/28/2023  07:09 PM    <DIR>          Desktop
06/27/2023  05:31 PM    <DIR>          Documents
06/28/2023  07:04 PM    <DIR>          Downloads
03/23/2023  05:04 AM    <DIR>          Favorites
03/23/2023  05:04 AM    <DIR>          Links
03/23/2023  05:04 AM    <DIR>          Music
05/20/2024  04:10 PM             1,310,720 NTUSER.bak
08/01/2024  03:26 PM             1,310,720 NTUSER.DAT
03/23/2023  03:54 AM    <DIR>          OneDrive
05/17/2023  04:45 PM    <DIR>          Pictures
03/23/2023  05:04 AM    <DIR>          Saved Games
03/23/2023  05:04 AM    <DIR>          Searches
03/31/2023  03:47 AM    <DIR>          Videos
                2 File(s)            2,621,440 bytes
               16 Dir(s)  13,968,850,944 bytes free
```

3. Notice that the *CLI* directory exists in the current working directory. Now let's change to the **CLI** directory by entering the **cd** command below.

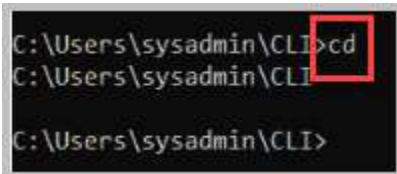
```
cd CLI
```



```
C:\Users\sysadmin>cd CLI
C:\Users\sysadmin\CLI>
```


4. Use the **cd** command to print the current working directory, ensuring that you are in the *CLI* directory.

```
cd
```



```
C:\Users\sysadmin\CLI>cd
C:\Users\sysadmin\CLI>
```

5. Create another directory, called **Meeting**, by entering the command below.

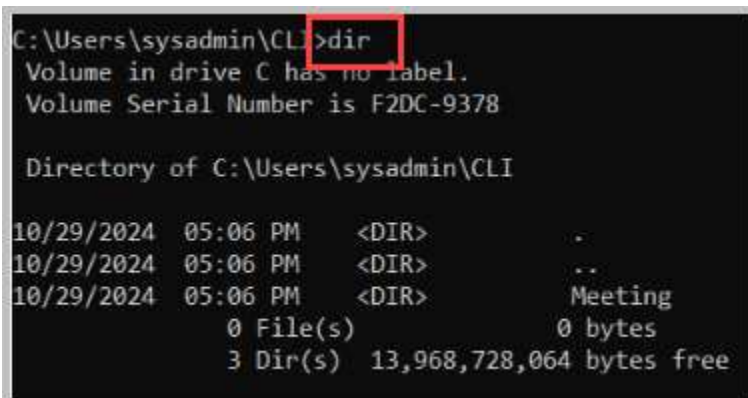
```
md Meeting
```



```
C:\Users\sysadmin\CLI>md Meeting
C:\Users\sysadmin\CLI>
```

6. Enter the **dir** command to verify that the *Meeting* directory has been created.

```
dir
```



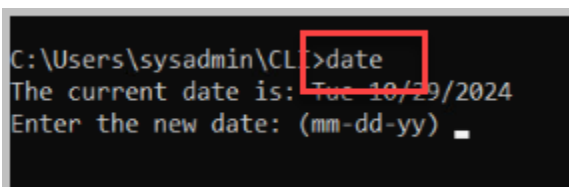
```
C:\Users\sysadmin\CLI>dir
Volume in drive C has no label.
Volume Serial Number is F2DC-9378

Directory of C:\Users\sysadmin\CLI

10/29/2024  05:06 PM    <DIR>          .
10/29/2024  05:06 PM    <DIR>          ..
10/29/2024  05:06 PM    <DIR>          Meeting
               0 File(s)                0 bytes
               3 Dir(s) 13,968,728,064 bytes free
```

7. Now, let's create files with the command prompt. The first method you will use to create a file is to redirect the output of a command to it. This is called *redirection*, and you use the *greater than* ">" redirector symbol. At the command prompt, type the command below. The output of this command will be the data in the calendar file you create.

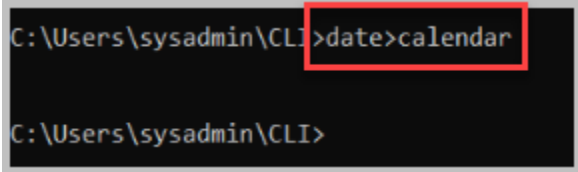
```
date
```



```
C:\Users\sysadmin\CLI>date
The current date is: Tue 10/29/2024
Enter the new date: (mm-dd-yy)
```


8. Press **CTRL+C** to exit the date command.
9. Now type the command below to redirect the data from the *date* command to a file named **calendar** in the current working directory. You will need to press Enter again to get the prompt back.

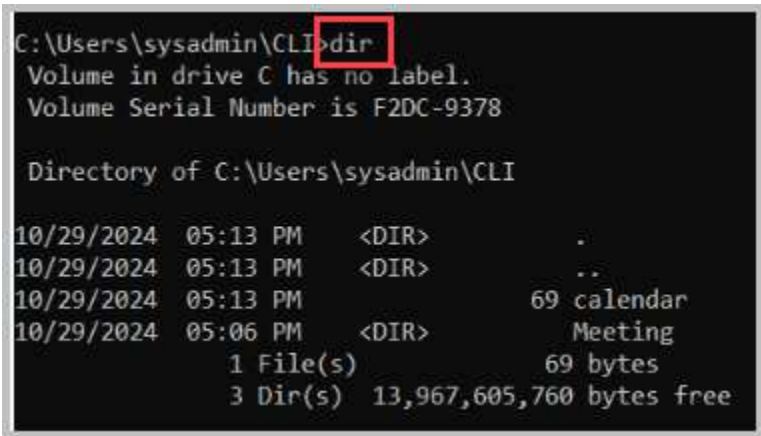
```
date>calendar
```



```
C:\Users\sysadmin\CLI>date>calendar  
  
C:\Users\sysadmin\CLI>
```

10. Verify that the calendar file exists by entering the **dir** command.

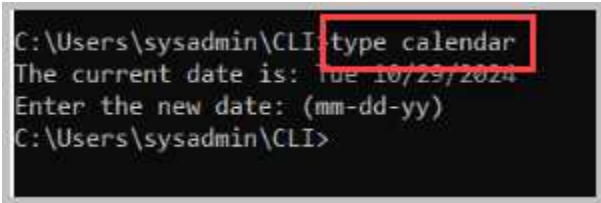
```
dir
```



```
C:\Users\sysadmin\CLI>dir  
Volume in drive C has no label.  
Volume Serial Number is F2DC-9378  
  
Directory of C:\Users\sysadmin\CLI  
  
10/29/2024 05:13 PM <DIR> .  
10/29/2024 05:13 PM <DIR> ..  
10/29/2024 05:13 PM 69 calendar  
10/29/2024 05:06 PM <DIR> Meeting  
1 File(s) 69 bytes  
3 Dir(s) 13,967,605,760 bytes free
```

11. Enter the command below to view the contents of the calendar file.

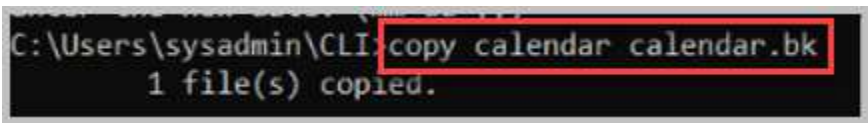
```
type calendar
```



```
C:\Users\sysadmin\CLI>type calendar  
The current date is: Tue 10/29/2024  
Enter the new date: (mm-dd-yy)  
C:\Users\sysadmin\CLI>
```

12. Enter the command below to create a backup of the calendar file named **calendar.bk**.

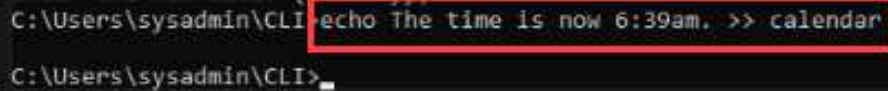
```
copy calendar calendar.bk
```



```
C:\Users\sysadmin\CLI>copy calendar calendar.bk  
1 file(s) copied.
```

13. The option to add additional information to either the calendar file or the calendar.bk file is available by using the echo command. Enter the command below to append the information to the calendar file.

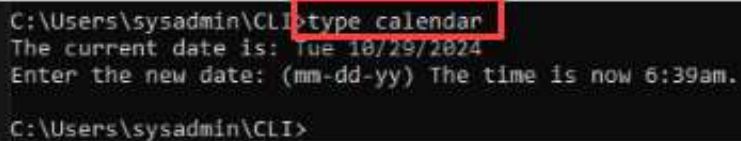
```
echo The time is now 6:39am. >> calendar
```



```
C:\Users\sysadmin\CLI>echo The time is now 6:39am. >> calendar
C:\Users\sysadmin\CLI>
```

14. Enter the command below to view the contents of the updated calendar file. Notice the updated information is now listed in the calendar file.

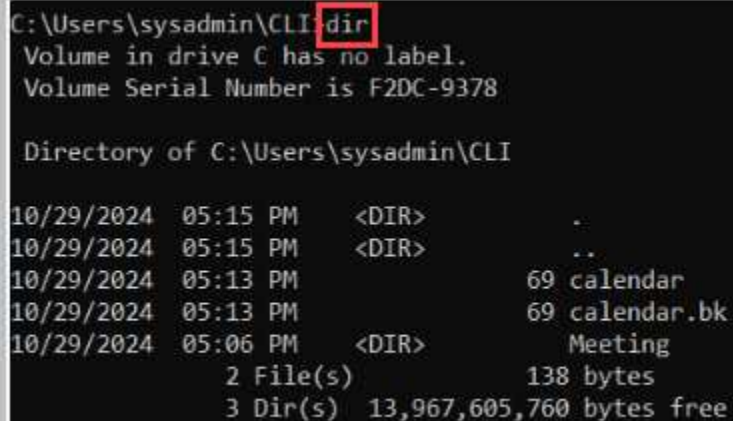
```
type calendar
```



```
C:\Users\sysadmin\CLI>type calendar
The current date is: Tue 10/29/2024
Enter the new date: (mm-dd-yy) The time is now 6:39am.
C:\Users\sysadmin\CLI>
```

15. Verify that the file was copied successfully by using the **dir** command.

```
dir
```



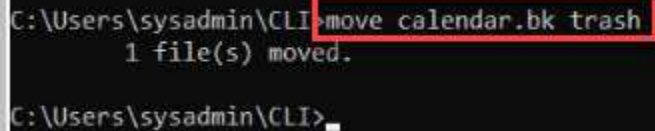
```
C:\Users\sysadmin\CLI>dir
Volume in drive C has no label.
Volume Serial Number is F2DC-9378

Directory of C:\Users\sysadmin\CLI

10/29/2024  05:15 PM    <DIR>          .
10/29/2024  05:15 PM    <DIR>          ..
10/29/2024  05:13 PM                69 calendar
10/29/2024  05:13 PM                69 calendar.bk
10/29/2024  05:06 PM    <DIR>          Meeting
                2 File(s)            138 bytes
                3 Dir(s)  13,967,605,760 bytes free
```

16. Type the **move** command below to rename the **calendar.bk** file as “**trash**” followed by pressing the **Enter** key.

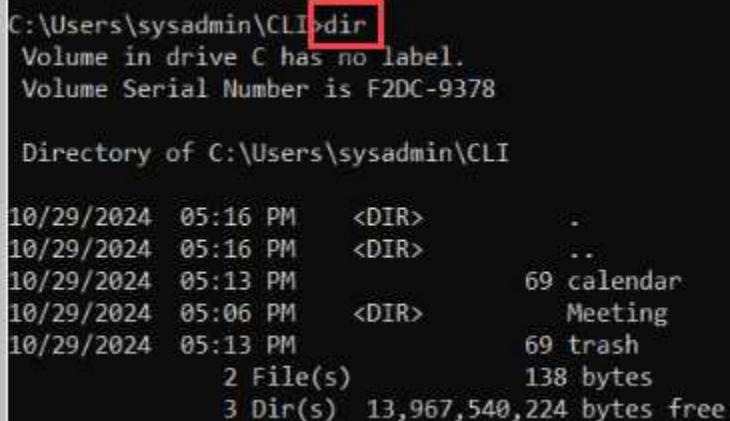
```
move calendar.bk trash
```



```
C:\Users\sysadmin\CLI>move calendar.bk trash
1 file(s) moved.
C:\Users\sysadmin\CLI>
```

17. Enter the **dir** command to see the result of the previous command. Notice the **calendar.bk** file was renamed to **trash**.

```
dir
```



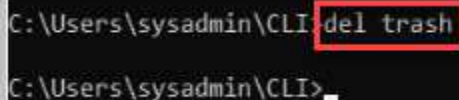
```
C:\Users\sysadmin\CLI>dir
Volume in drive C has no label.
Volume Serial Number is F2DC-9378

Directory of C:\Users\sysadmin\CLI

10/29/2024  05:16 PM    <DIR>          .
10/29/2024  05:16 PM    <DIR>          ..
10/29/2024  05:13 PM             69 calendar
10/29/2024  05:06 PM    <DIR>          Meeting
10/29/2024  05:13 PM             69 trash
                2 File(s)              138 bytes
                3 Dir(s)  13,967,540,224 bytes free
```

18. Enter the **del** command below to remove the **trash** file from the **CLI** directory.

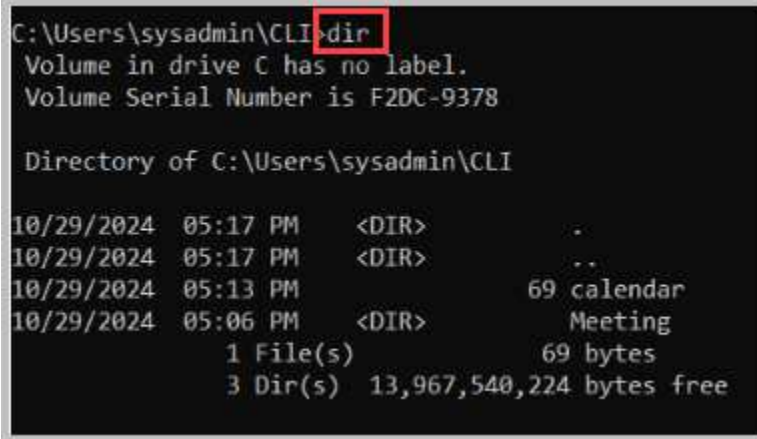
```
del trash
```



```
C:\Users\sysadmin\CLI>del trash
C:\Users\sysadmin\CLI>
```

19. Verify that the **trash** file was removed from the *CLI* directory by entering the **dir** command.

dir



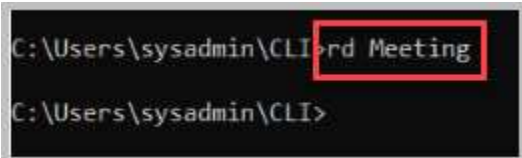
```
C:\Users\sysadmin\CLI>dir
Volume in drive C has no label.
Volume Serial Number is F2DC-9378

Directory of C:\Users\sysadmin\CLI

10/29/2024  05:17 PM    <DIR>          .
10/29/2024  05:17 PM    <DIR>          ..
10/29/2024  05:13 PM                69 calendar
10/29/2024  05:06 PM    <DIR>          Meeting
                1 File(s)                69 bytes
                3 Dir(s)  13,967,540,224 bytes free
```

20. Type the **rd** command below to delete the **Meeting** directory (folder).

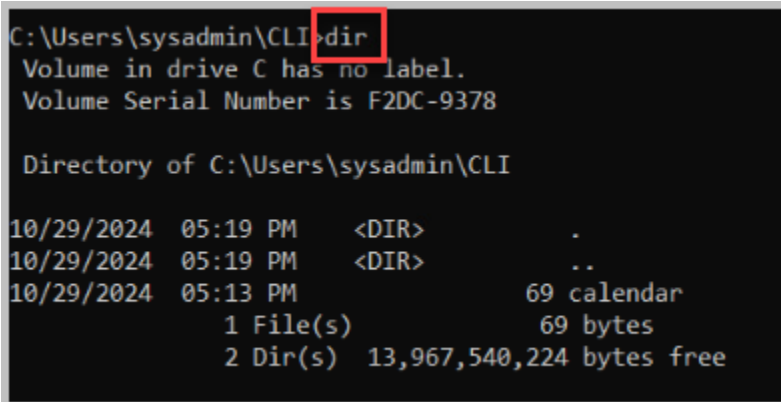
rd Meeting



```
C:\Users\sysadmin\CLI>rd Meeting
C:\Users\sysadmin\CLI>
```

21. Verify the *Meeting* directory has been removed by entering the **dir** command.

dir



```
C:\Users\sysadmin\CLI>dir
Volume in drive C has no label.
Volume Serial Number is F2DC-9378

Directory of C:\Users\sysadmin\CLI

10/29/2024  05:19 PM    <DIR>          .
10/29/2024  05:19 PM    <DIR>          ..
10/29/2024  05:13 PM                69 calendar
                1 File(s)                69 bytes
                2 Dir(s)  13,967,540,224 bytes free
```

22. Type **exit** followed by pressing the **Enter** key to quit the command prompt.

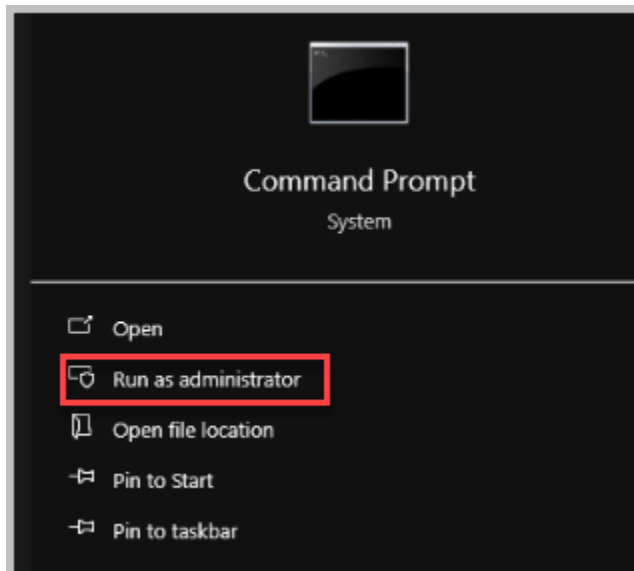
1.1.4 Command Prompt with Administrative Privileges

Using the *Windows 10 operating system*, you will work in *Command Prompt* with elevated privileges. Running certain commands in *Windows* requires elevated privileges, even if you are logged in with an account that has administrative rights. This is for security reasons. In this exercise, you will access the command prompt as *Administrator* to execute some of these commands.

1. In the search box of the taskbar, type **Command Prompt**.



2. Select the option to **Run as administrator**.



3. When prompted with a *User Account Control* message, click **Yes** to continue.

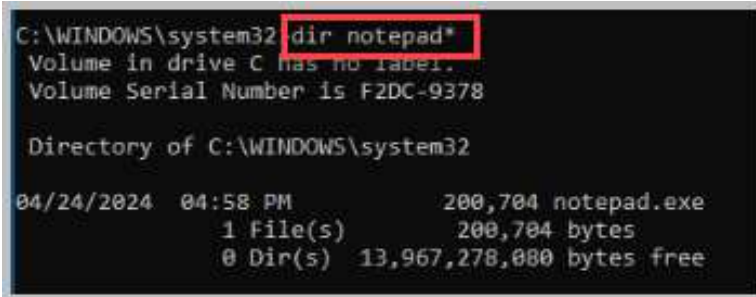


- Once the command prompt shell appears, notice the header in the title bar signaling that the prompt is opened with administrator privileges. Enter the **dir** command, followed by pressing the **Enter** key to list files and folders available in the current working directory.

```
dir
```

- You'll notice that a lot of output is given due to the number of files and folders present in the `C:\WINDOWS\system32` directory, which is the default directory when opening the command prompt with administrative privileges. Try to see if there are any files associated with the *Notepad* application by entering the command below using a wildcard parameter.

```
dir notepad*
```



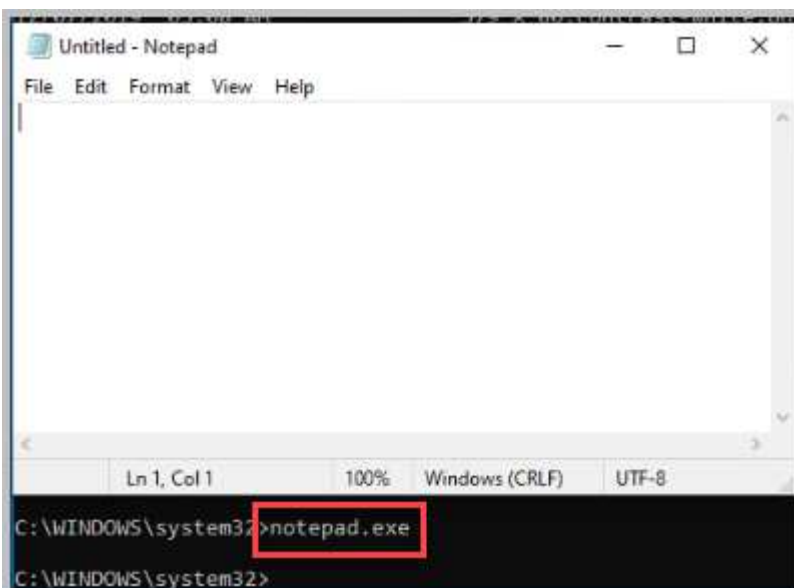
```
C:\WINDOWS\system32>dir notepad*
Volume in drive C has no label.
Volume Serial Number is F2DC-9378

Directory of C:\WINDOWS\system32

04/24/2024  04:58 PM                200,704 notepad.exe
               1 File(s)                200,704 bytes
               0 Dir(s) 13,967,278,080 bytes free
```

- Notice that a *notepad.exe* executable file is present. Enter the command below, followed by pressing the **Enter** key. This command will launch the **Notepad** application directly from the shell as long as the program's name is in the directory where it is located. Since we are already in `C:\WINDOWS\system32`, the *notepad.exe* executable file is in the current working directory by default.

```
notepad.exe
```



7. Notice the *Notepad* application launches. Change focus to the **command prompt** and enter the **tasklist** command to display a list of currently running processes.

```
tasklist
```

```
C:\WINDOWS\system32>tasklist
```

Image Name	PID	Session Name	Session#	Mem Usage
System Idle Process	0	Services	0	8 K
System	4	Services	0	136 K
Registry	92	Services	0	19,972 K
smss.exe	364	Services	0	1,184 K
csrss.exe	472	Services	0	5,480 K
wininit.exe	572	Services	0	7,108 K
csrss.exe	580	Console	1	5,476 K
winlogon.exe	664	Console	1	11,848 K
services.exe	700	Services	0	10,192 K
lsass.exe	716	Services	0	17,200 K
fontdrvhost.exe	804	Services	0	4,640 K
svchost.exe	820	Services	0	27,864 K
fontdrvhost.exe	828	Console	1	5,860 K
svchost.exe	940	Services	0	15,012 K
svchost.exe	988	Services	0	8,408 K

8. Locate **notepad.exe** from the list and notice the information provided, such as PID and *Mem Usage*. Enter the command below to terminate the Notepad process. This lets you manage a task and manually kill a rogue process.

```
taskkill /IM notepad.exe
```

```
C:\WINDOWS\system32>taskkill /IM notepad.exe
SUCCESS: Sent termination signal to the process "notepad.exe" with PID 7252.
C:\WINDOWS\system32>_
```


9. Enter the **help** command below to learn what the */M* switch means.

```
taskkill /?
```

```
C:\WINDOWS\system32>taskkill /?

TASKKILL [/S system [/U username [/P [password]]]]
{ [/FI filter] [/PID processid | /IM imagename] } [/T] [/F]

Description:
  This tool is used to terminate tasks by process id (PID) or image name.

Parameter List:
  /S    system          Specifies the remote system to connect to.
  /U    [domain\]user   Specifies the user context under which the
                        command should execute.
  /P    [password]      Specifies the password for the given user
                        context. Prompts for input if omitted.
  /FI   filter          Applies a filter to select a set of tasks.
                        Allows "*" to be used. ex. imagename eq acme*
  /PID  processid       Specifies the PID of the process to be terminated.
                        Use Tasklist to get the PID.
  /IM   imagename       Specifies the image name of the process
                        to be terminated. Wildcard '*' can be used
                        to specify all tasks or image names.
```

10. Enter the **tasklist** command once more to verify that *notepad.exe* is no longer available.

```
tasklist
```

```
C:\WINDOWS\system32>tasklist

Image Name                PID Session Name        Session#    Mem Usage
-----
System Idle Process        0 Services            0             8 K
System                     4 Services            0            136 K
Registry                   92 Services            0          19,972 K
smss.exe                   364 Services            0           1,184 K
csrss.exe                  472 Services            0           5,480 K
wininit.exe                572 Services            0           7,108 K
csrss.exe                  580 Console             1           5,476 K
winlogon.exe               664 Console             1          11,848 K
services.exe              700 Services            0          10,192 K
lsass.exe                  716 Services            0          17,200 K
fontdrvhost.exe           804 Services            0           4,640 K
svchost.exe                820 Services            0          27,864 K
fontdrvhost.exe           828 Console             1           5,860 K
svchost.exe               940 Services            0          15,012 K
svchost.exe              988 Services            0           8,408 K
```


11. Enter the help command for the **shutdown** command. The *shutdown* command can be used to shut down, restart, log off, or hibernate your *Windows* machine.

```
shutdown /?
```

```
C:\WINDOWS\system32 shutdown /?
Usage: shutdown [/i | /l | /s | /sg | /r | /g | /a | /p | /h | /e | /o] [/hybrid] [/soft] [/fw] [/f]
[/m \\computer] [/t xxx] [/d [p|u:]xx:yy [/c "comment"]]
```

No args	Display help. This is the same as typing /?.
/?	Display help. This is the same as not typing any options.
/i	Display the graphical user interface (GUI). This must be the first option.
/l	Log off. This cannot be used with /m or /d options.
/s	Shutdown the computer.
/sg	Shutdown the computer. On the next boot, if Automatic Restart Sign-On is enabled, automatically sign in and lock last interactive user. After sign in, restart any registered applications.
/r	Full shutdown and restart the computer.
/g	Full shutdown and restart the computer. After the system is rebooted, if Automatic Restart Sign-On is enabled, automatically sign in and lock last interactive user. After sign in, restart any registered applications.
/a	Abort a system shutdown. This can only be used during the time-out period.
/p	Turn off the local computer with no time-out or warning. Can be used with /d and /f options.
/h	Hibernate the local computer. Can be used with the /f option.
/hybrid	Performs a shutdown of the computer and prepares it for fast startup. Must be used with /s option.
/fw	Combine with a shutdown option to cause the next boot to go to the firmware user interface.
/e	Document the reason for an unexpected shutdown of a computer.
/o	Go to the advanced boot options menu and restart the computer. Must be used with /r option.
/m \\computer	Specify the target computer.
/t xxx	Set the time-out period before shutdown to xxx seconds. The valid range is 0-315360000 (10 years), with a default of 30. If the timeout period is greater than 0, the /f parameter is implied.
/c "comment"	Comment on the reason for the restart or shutdown. Maximum of 512 characters allowed.
/f	Force running applications to close without forewarning users. The /f parameter is implied when a value greater than 0 is specified for the /t parameter.
/d [p u:]xx:yy	Provide the reason for the restart or shutdown. p indicates that the restart or shutdown is planned. u indicates that the reason is user defined. If neither p nor u is specified the restart or shutdown is unplanned. xx is the major reason number (positive integer less than 256).

12. Enter the command below to log off from the *Win10-OS* system.

```
shutdown /l
```

2 Navigating the Command Line Interface in Linux

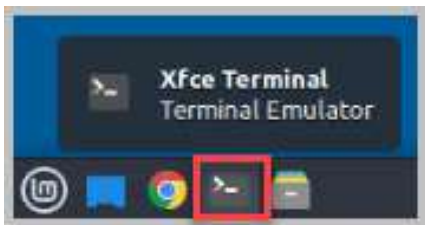
While most *Linux* distros offer a GUI, many Linux users prefer the CLI because it is a quick way to perform tasks. In this exercise, you will use basic *Linux* commands to carry out tasks on the system.

2.1 Using the CLI in Lubuntu

1. Open a console connection to the **Linux-OS** system.



2. In the taskbar of Linux-OS, click on the **XFceTerminal** button to open the terminal window, which in comparison to *Windows*, is similar to the command prompt with regards to using a command-line interface.



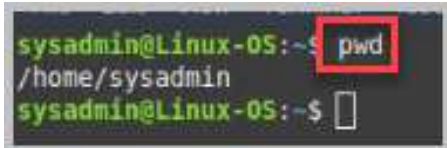
3. Leave the terminal open to continue with the next task.

2.1.1 View the Contents of a Directory

In *Linux*, when you open the *CLI*, the prompt shows the current working directory, similar to *Windows*. It is very important to know the directory you are currently working in.

1. At the terminal, type the command below, followed by pressing the **Enter** key. This command displays the current working directory path.

```
pwd
```



```
sysadmin@Linux-OS:~$ pwd
/home/sysadmin
sysadmin@Linux-OS:~$
```



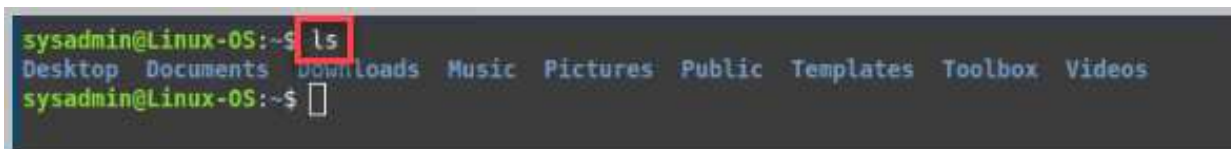
When typing commands in the terminal, typically at the end of every typed command you press the **Enter** key to initiate the command. Also, if you accidentally misspelled a command, as a shortcut, you could press the **up arrow** to bring back the previous commands you've entered.



To get help at any place in the terminal, type **help** and a list of all available commands will display. If you need help for a specific command type **man <command>**.

2. Type the **ls** command, followed by pressing the **Enter** key. This command displays a list of files and directories inside the current working directory.

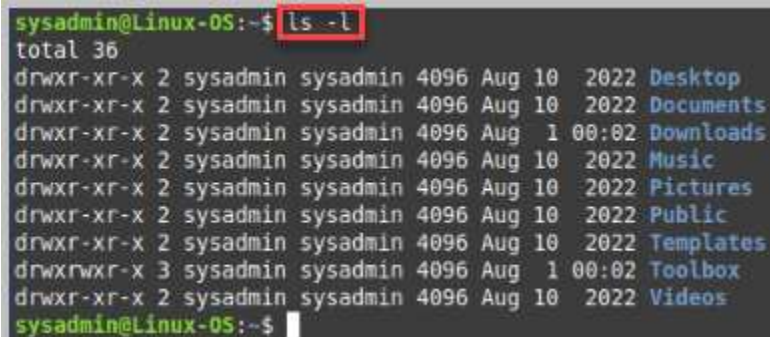
```
ls
```



```
sysadmin@Linux-OS:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Toolbox Videos
sysadmin@Linux-OS:~$
```

3. Enter the command below to display a detailed list of files and directories.

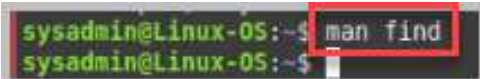
```
ls -l
```



```
sysadmin@Linux-OS:~$ ls -l
total 36
drwxr-xr-x 2 sysadmin sysadmin 4096 Aug 10 2022 Desktop
drwxr-xr-x 2 sysadmin sysadmin 4096 Aug 10 2022 Documents
drwxr-xr-x 2 sysadmin sysadmin 4096 Aug 1 00:02 Downloads
drwxr-xr-x 2 sysadmin sysadmin 4096 Aug 10 2022 Music
drwxr-xr-x 2 sysadmin sysadmin 4096 Aug 10 2022 Pictures
drwxr-xr-x 2 sysadmin sysadmin 4096 Aug 10 2022 Public
drwxr-xr-x 2 sysadmin sysadmin 4096 Aug 10 2022 Templates
drwxr-xr-x 2 sysadmin sysadmin 4096 Aug 1 00:02 Toolbox
drwxr-xr-x 2 sysadmin sysadmin 4096 Aug 10 2022 Videos
sysadmin@Linux-OS:~$
```

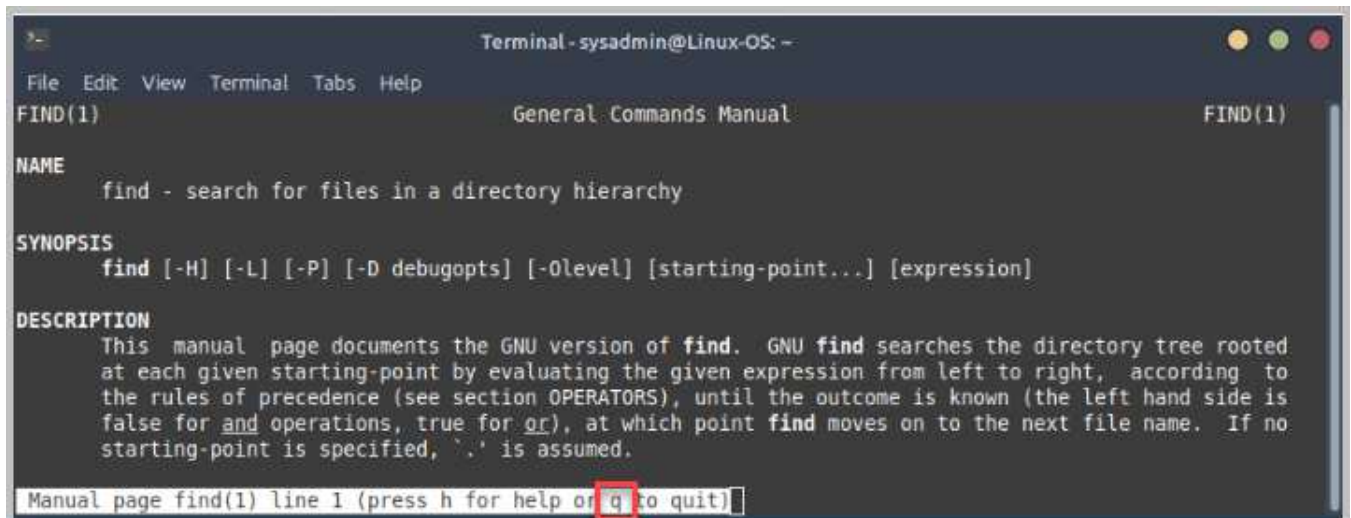
4. An alternative method to display a comprehensive list of files and directories involves using the find command. To fully understand its capabilities, it's advisable to consult the manual page for find. You can access this by entering the following command, which will bring up the manual page for the find command.

```
man find
```



```
sysadmin@Linux-OS:~$ man find
sysadmin@Linux-OS:~$
```

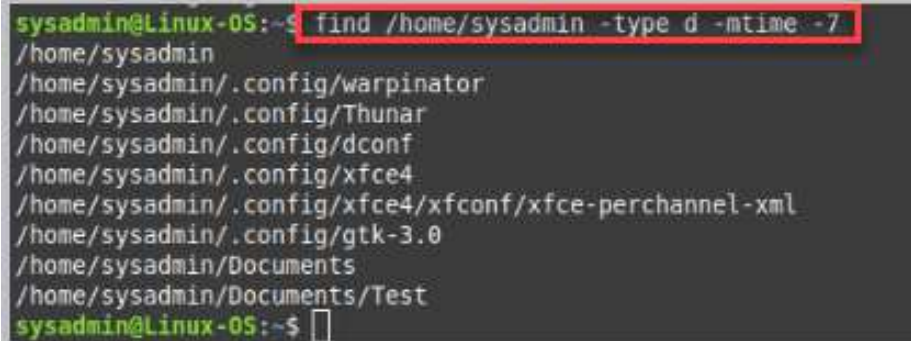
5. The output from using this command is shown below. To exit the Manual Page, type **q** to quit.



```
Terminal - sysadmin@Linux-OS: -
File Edit View Terminal Tabs Help
FIND(1) General Commands Manual FIND(1)
NAME
  find - search for files in a directory hierarchy
SYNOPSIS
  find [-H] [-L] [-P] [-D debugopts] [-O level] [starting-point...] [expression]
DESCRIPTION
  This manual page documents the GNU version of find. GNU find searches the directory tree rooted
  at each given starting-point by evaluating the given expression from left to right, according to
  the rules of precedence (see section OPERATORS), until the outcome is known (the left hand side is
  false for and operations, true for or), at which point find moves on to the next file name. If no
  starting-point is specified, '.' is assumed.
Manual page find(1) line 1 (press h for help or q to quit)
```

- As there are numerous options with the find command, enter the find command below to display the directories Modification Time. This command finds directories in the /home/sysadmin directory that have been modified in the last 7 days.

```
find /home/sysadmin -type d -mtime -7
```



```
sysadmin@Linux-OS:~$ find /home/sysadmin -type d -mtime -7
/home/sysadmin
/home/sysadmin/.config/warpinator
/home/sysadmin/.config/Thunar
/home/sysadmin/.config/dconf
/home/sysadmin/.config/xfce4
/home/sysadmin/.config/xfce4/xfconf/xfce-perchannel-xml
/home/sysadmin/.config/gtk-3.0
/home/sysadmin/Documents
/home/sysadmin/Documents/Test
sysadmin@Linux-OS:~$
```

- Leave the terminal open to continue with the next task.

2.1.2 Working with Directories

- At the terminal prompt, enter the command below to move into the **Documents** directory. The **cd** command allows you to change between directories.

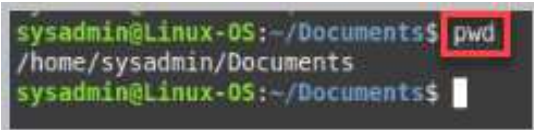
```
cd Documents
```



```
sysadmin@Linux-OS:~$ cd Documents
sysadmin@Linux-OS:~/Documents$
```

- Enter the **pwd** command to identify your current working directory. Notice you are now in the /home/sysadmin/Documents directory.

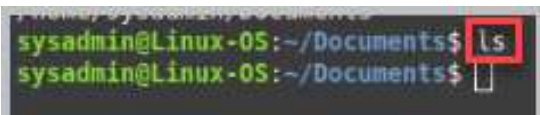
```
pwd
```



```
sysadmin@Linux-OS:~/Documents$ pwd
/home/sysadmin/Documents
sysadmin@Linux-OS:~/Documents$
```

- Enter the **ls** command to view the contents of the *Documents* directory. Notice no files or directories are present here.

```
ls
```



```
sysadmin@Linux-OS:~/Documents$ ls
sysadmin@Linux-OS:~/Documents$
```

4. Leave the terminal open to continue with the next task.

2.1.3 Creating Directories and Files

1. At the terminal, enter the **mkdir** command below to create a new directory called **Test** in the current working directory.

```
mkdir Test
```



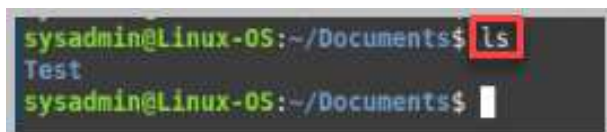
```
sysadmin@Linux-OS:~/Documents$ mkdir Test
sysadmin@Linux-OS:~/Documents$
```



The command-line is not interactive for many commands. If you do not get an error message, you may assume your command succeeded. You will not be asked if you are sure you want to continue or be given feedback about the action taken, so it is important to verify the results of your actions.

2. Verify that the new *Test* directory exists by entering the **ls** command.

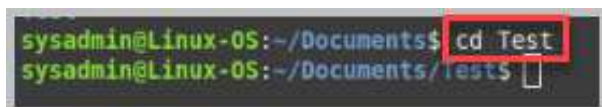
```
ls
```



```
sysadmin@Linux-OS:~/Documents$ ls
Test
sysadmin@Linux-OS:~/Documents$
```

3. Enter the **cd** command below to change to the **Test** directory.

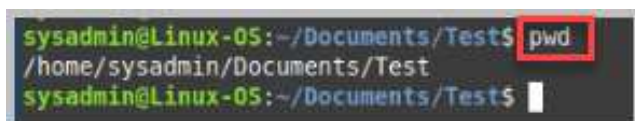
```
cd Test
```



```
sysadmin@Linux-OS:~/Documents$ cd Test
sysadmin@Linux-OS:~/Documents/Test$
```

4. Enter the **pwd** command to verify your current working directory.

```
pwd
```



```
sysadmin@Linux-OS:~/Documents/Test$ pwd
/home/sysadmin/Documents/Test
sysadmin@Linux-OS:~/Documents/Test$
```


5. Enter the command below to create a new **Redirect** directory in *Test*.

```
mkdir Redirect
```



```
sysadmin@Linux-OS:~/Documents/Test$ mkdir Redirect
sysadmin@Linux-OS:~/Documents/Test$
```

6. Enter the **ls** command to verify that the *Redirect* directory has been successfully created.

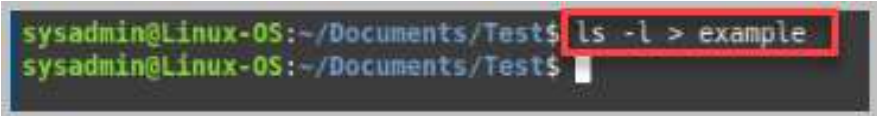
```
ls
```



```
sysadmin@Linux-OS:~/Documents/Test$ ls
Redirect
sysadmin@Linux-OS:~/Documents/Test$
```

7. Now, it's time to create files. The first method you will use to create a file is to send the output of a command to a file. This is called *redirection*, and you use the *greater than* ">" or *pipe* "|" redirector symbol to redirect output from one command into another. This command also lets you chain commands together into a more robust command. Enter the command below to redirect the output from the **ls** command to a file named **example**.

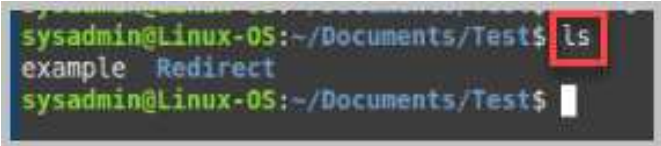
```
ls -l > example
```



```
sysadmin@Linux-OS:~/Documents/Test$ ls -l > example
sysadmin@Linux-OS:~/Documents/Test$
```

8. Verify that the *example* file exists by entering the **ls** command.

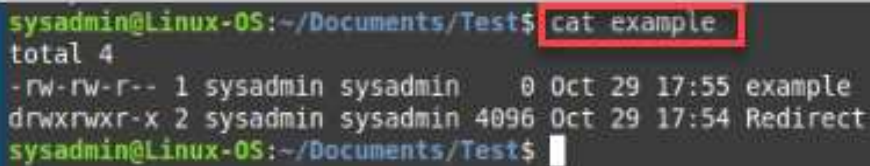
```
ls
```



```
sysadmin@Linux-OS:~/Documents/Test$ ls
example Redirect
sysadmin@Linux-OS:~/Documents/Test$
```

9. View the contents of the **example** file by using the **cat** command. Enter the command below. Notice that the output from the **ls -l** command is copied in this file.

```
cat example
```



```
sysadmin@Linux-OS:~/Documents/Test$ cat example
total 4
-rw-rw-r-- 1 sysadmin sysadmin  0 Oct 29 17:55 example
drwxrwxr-x 2 sysadmin sysadmin 4096 Oct 29 17:54 Redirect
sysadmin@Linux-OS:~/Documents/Test$
```

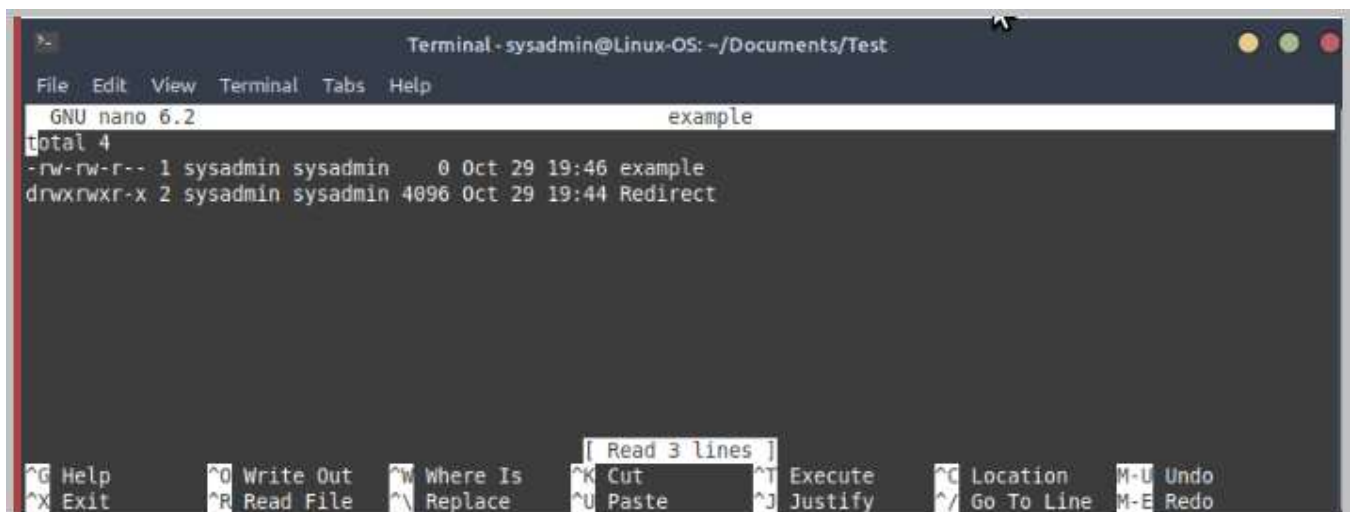
10. A secondary way of viewing the contents of the **example** file is to use the **nano** command. *Nano* is a command-line text editor known for its simplicity and ease of use. Enter the command below and select **enter**.

```
nano example
```



```
sysadmin@Linux-OS:~/Documents/Test$ nano example
```

11. Notice that the output from the *nano* command matches the output from the *cat* command. To exit the nano text editor, select **Ctrl + x** from your keyboard. This will exit the *Nano Editor*.



12. Make a copy of the **example** file and call it **example.bk**. Enter the command below.

```
cp example example.bk
```



```
sysadmin@Linux-OS:~/Documents/Tests$ cp example example.bk
sysadmin@Linux-OS:~/Documents/Tests$
```

13. Verify that the file was created successfully using the **ls** command.

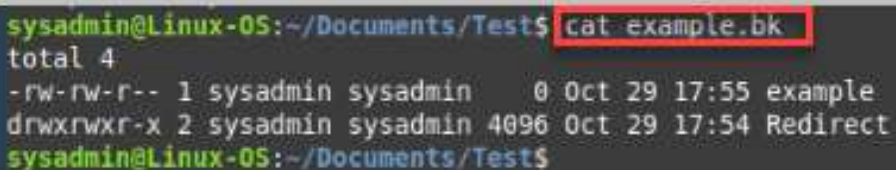
```
ls
```



```
sysadmin@Linux-OS:~/Documents/Tests$ ls
example example.bk Redirect
sysadmin@Linux-OS:~/Documents/Tests$
```

14. Verify that the contents of the file were copied successfully by using the **cat** command. Enter the command below.

```
cat example.bk
```



```
sysadmin@Linux-OS:~/Documents/Tests$ cat example.bk
total 4
-rw-rw-r-- 1 sysadmin sysadmin 0 Oct 29 17:55 example
drwxrwxr-x 2 sysadmin sysadmin 4096 Oct 29 17:54 Redirect
sysadmin@Linux-OS:~/Documents/Tests$
```

15. Enter the **mv** command below to rename the *example.bk* file to **sample**.

```
mv example.bk sample
```



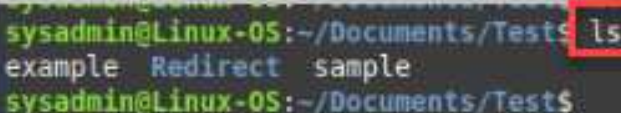
```
sysadmin@Linux-OS:~/Documents/Tests$ mv example.bk sample
sysadmin@Linux-OS:~/Documents/Tests$
```



Even though the *mv* command is used to move files, it can also be used to rename files.

16. Verify that the file has been renamed successfully by entering the **ls** command.

```
ls
```



```
sysadmin@Linux-OS:~/Documents/Tests$ ls
example Redirect sample
sysadmin@Linux-OS:~/Documents/Tests$
```

17. Enter the command below to delete the **sample** file.

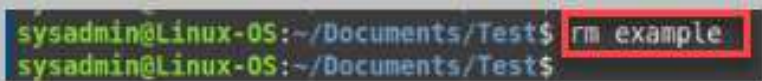
```
rm sample
```



```
sysadmin@Linux-OS:~/Documents/Tests$ rm example
```

18. Remove the **example** file using the **rm** command.

```
rm example
```



```
sysadmin@Linux-OS:~/Documents/Tests$ rm example
sysadmin@Linux-OS:~/Documents/Tests$
```

19. Enter the **ls** command to verify that both the *example* and *sample* files have been removed from the *Test* directory.

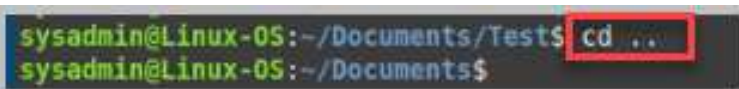
```
ls
```



```
sysadmin@Linux-OS:~/Documents/Tests$ ls
Redirect sample
sysadmin@Linux-OS:~/Documents/Tests$
```

20. Use the **cd** command to go back to the **Documents** directory as your working directory.

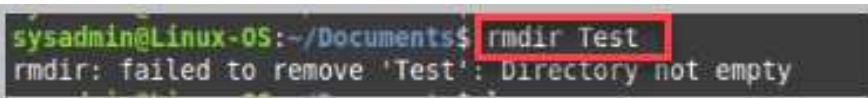
```
cd ..
```



```
sysadmin@Linux-OS:~/Documents/Tests$ cd ..
sysadmin@Linux-OS:~/Documents$
```

21. Remove the **Test** directory by entering the command below.

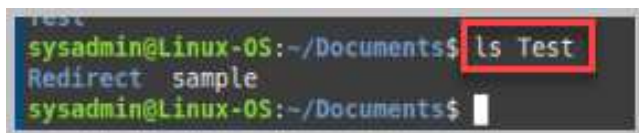
```
rmdir Test
```



```
sysadmin@Linux-OS:~/Documents$ rmdir Test
rmdir: failed to remove 'Test': Directory not empty
```

22. Notice the message that appears. Run the `ls` command below to see what other objects in the `Test` directory are preventing us from deleting it.

```
ls Test
```



```
sysadmin@Linux-OS:~/Documents$ ls Test
Redirect sample
sysadmin@Linux-OS:~/Documents$
```

23. Notice there is another directory that we created called *Redirect*. To remove a directory with subdirectories, we need to use additional parameters along with the `rm` command. Enter the command below.

```
rm -r Test
```



```
sysadmin@Linux-OS:~/Documents$ rm -r Test
sysadmin@Linux-OS:~/Documents$
```



The `-r` option for the `rm` command is used to remove directories and their contents recursively.

24. Enter the `ls` command to verify that the *Test* directory is gone.

```
ls
```



```
sysadmin@Linux-OS:~/Documents$ ls
sysadmin@Linux-OS:~/Documents$
```

25. Type `exit` followed by pressing the **Enter** key to quit the terminal.
26. The lab is now complete; you may end the reservation.